

PORT ALFRED, South Africa

WMO: 688430

Lat: 33.559S

Lon: 26.881E

Elev: 84

StdP: 100.32

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	6.6	-1.4	3.4	18.7	0.2	3.9	16.1	11.1	16.5	9.8	15.1	2.7	280	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.7	29.6	19.5	27.7	19.8	26.2	19.8	22.8	26.7	22.1	25.8	21.4	24.9	5.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.6	16.4	24.6	20.8	15.7	24.2	20.1	14.9	23.7	68.0	26.7	65.1	25.8	62.7	25.1	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.3	8.1	3.4	34.4	1.8	2.1	2.1	35.9	1.0	37.1	0.0	38.2	-1.3	39.7
			1.7	24.0	1.1	0.4	0.9	24.3	0.3	24.5	-0.4	24.8	-1.2	25.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	21.2	21.5	20.4	18.2	16.7	14.9	14.7	14.7	15.6	16.9	18.0	19.9
	DBStd	3.38	1.94	2.14	2.27	2.50	2.42	2.69	2.98	2.53	2.16	2.30	2.04	2.07
	HDD10.0	2	0	0	0	0	1	1	1	1	0	0	0	0
	HDD18.3	632	2	2	6	30	63	111	122	117	87	56	29	8
	CDD10.0	2818	348	322	323	247	208	147	146	147	168	214	241	307
	CDD18.3	407	91	90	71	27	12	7	9	5	4	12	21	57
	CDH23.3	1477	311	316	212	114	78	64	76	42	27	49	36	151
	CDH26.7	284	42	54	45	31	19	15	18	14	7	11	3	24
(14) Wind	WSAvg	3.6	4.0	3.8	3.4	3.2	3.1	3.5	3.4	3.4	3.7	4.0	4.0	4.0
(15) Precipitation	PrecAvg	632	41	60	63	53	38	53	44	69	57	77	62	43
	PrecMax	947	64	129	128	149	127	187	148	281	190	171	161	128
	PrecMin	460	20	24	12	5	7	5	2	14	9	22	17	8
	PrecStd	149	14	28	32	44	34	46	45	68	50	44	39	29
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	31.4	31.2	30.4	29.1	28.9	28.9	28.8	26.8	27.8	26.8	29.6
		MCWB	22.5	22.2	21.8	19.7	16.1	14.3	14.2	14.2	16.5	18.6	19.6	21.2
	2%	DB	27.6	28.0	27.4	26.2	25.2	25.1	25.4	23.4	22.7	23.9	24.2	26.4
		MCWB	21.9	21.9	21.0	19.0	16.6	13.3	14.1	14.5	16.8	18.1	18.8	20.6
	5%	DB	26.1	26.3	25.4	23.8	22.7	21.8	21.8	20.9	20.8	21.9	23.0	24.7
		MCWB	21.3	21.4	20.4	18.5	16.0	13.0	13.4	14.6	16.3	17.4	18.4	19.8
	10%	DB	24.9	25.1	24.1	22.1	20.8	19.6	19.5	19.2	19.4	20.7	22.0	23.6
		MCWB	20.5	20.9	20.0	17.9	15.8	13.6	13.2	14.3	15.5	16.7	17.9	19.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.8	23.6	23.6	21.6	19.5	17.6	17.8	17.9	19.1	20.5	21.2	22.3
		MCDB	26.7	27.7	27.9	25.8	24.2	19.7	21.5	21.7	22.5	25.8	25.1	27.2
	2%	WB	22.6	22.7	22.4	20.4	18.2	16.1	16.1	16.7	17.9	19.1	20.1	21.2
		MCDB	26.5	26.6	25.6	24.1	22.0	20.0	20.0	20.1	21.1	22.5	23.3	25.2
	5%	WB	21.9	22.0	21.3	19.5	17.3	15.0	15.1	15.8	17.0	18.1	19.1	20.5
		MCDB	25.5	25.7	24.4	22.9	20.7	19.6	19.6	18.9	19.9	21.1	22.2	23.9
	10%	WB	21.1	21.4	20.5	18.6	16.6	14.1	14.2	14.9	16.1	17.3	18.3	19.7
		MCDB	24.4	24.8	23.5	21.8	19.8	18.6	18.3	18.3	19.0	20.0	21.4	22.8
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	6.7	7.0	7.9	9.3	10.3	10.2	9.5	8.0	6.9	6.9	6.6
		MCDBR	9.0	8.8	9.3	11.5	13.2	14.4	14.6	13.0	10.7	9.7	8.8	8.6
	5% WB	MCWBR	4.9	4.4	4.7	5.8	6.5	6.2	6.5	6.9	6.3	5.7	5.4	5.0
		MCWBR	8.5	8.0	8.1	9.2	10.2	10.8	11.1	9.9	9.4	8.8	8.1	8.2
(40) Clear-Sky Solar Irradiance	taub	taub	0.397	0.398	0.383	0.357	0.341	0.326	0.330	0.374	0.397	0.408	0.388	0.391
		taud	2.441	2.446	2.471	2.512	2.527	2.539	2.525	2.392	2.324	2.331	2.424	2.446
	Ebn at Noon	Ebn at Noon	941	919	896	865	825	814	829	831	866	898	943	950
		Edh at Noon	121	116	106	91	79	74	78	101	121	129	122	121
(44) All-Sky Solar Radiation	RadAvg	6.66	6.09	5.08	3.94	3.06	2.73	2.91	3.66	4.74	5.58	6.44	6.88	(44)
	RadStd	0.34	0.31	0.39	0.21	0.13	0.14	0.17	0.23	0.32	0.46	0.36	0.43	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.35	+0.42	N/A	N/A	N/A	N/A	N/A	-85	+142	+45

Nomenclature: See separate page